uart_fw Utility
User Guide

Version: 2.01.14

Table of Content

Chapter 1. Introduction

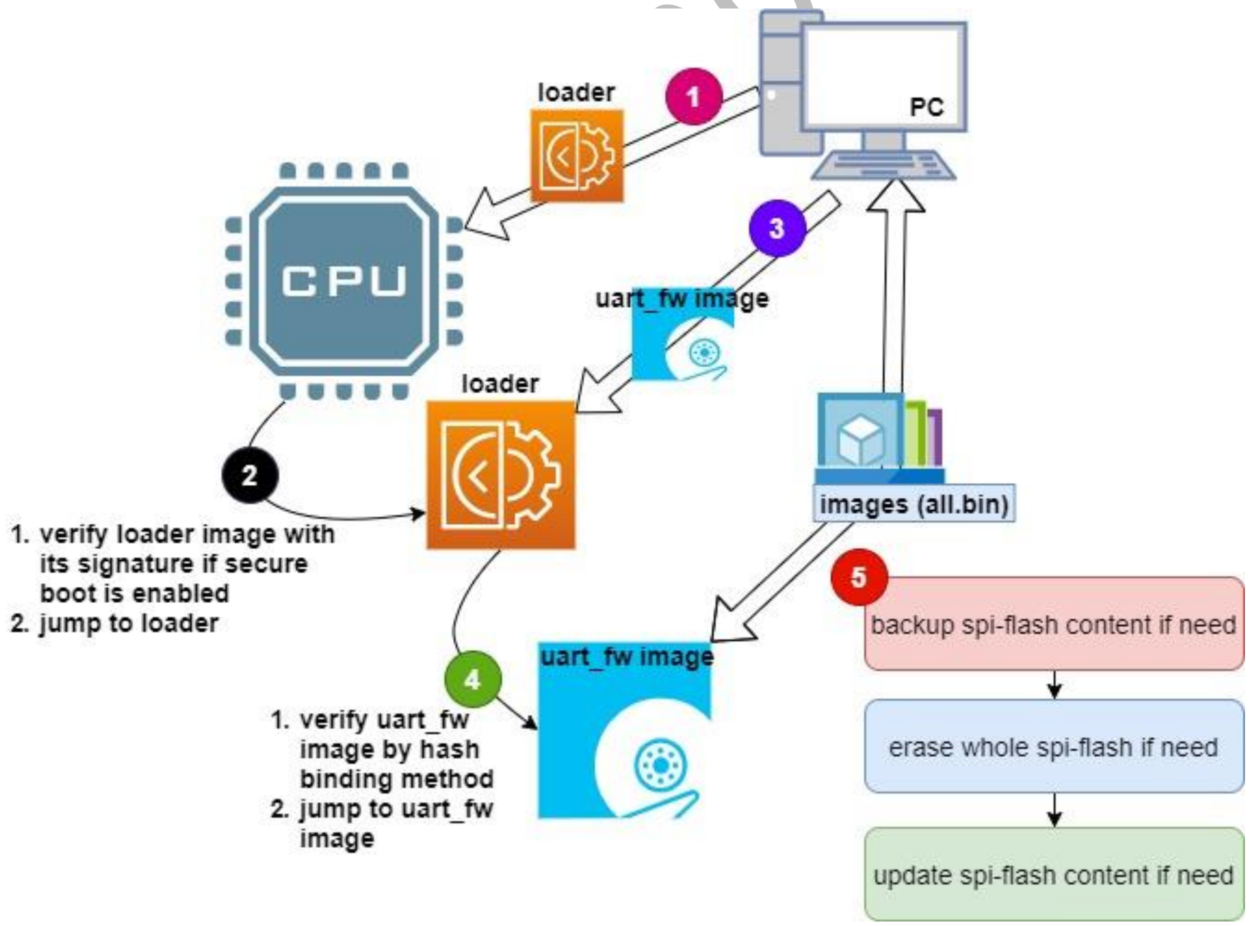
“boot from UART” is a new boot source created on/after AST2600, AST1030 or AST1060 series. Through UART port, BMC UART usually, MCU ROM can receive BMC image and locate it on SRAM, then ARM CPU can execute the boot up sequence with the content of SRAM.

uart_fw utilizes “boot from UART” path to read/write BMC SPI flash. **uart_fw** is a python tool designed for engineer development stage and can be executed on Windows/Linux X86-64 PC environment. Since “boot from UART” path is also protected by BMC secure boot, remember to sign your loader image before executing **uart_fw** utility when secure boot is enabled on your platform.

WARNING:

This utility is only for engineers to update the firmware for development purpose. It is not a commercialized software product and ASPEED has not done compatibility/reliability stress test. Please do not use this utility for any mass production purpose.

Chapter 2. uart_fw / uart_fw_gui Working Diagram



The following are roughly uart_fw execution steps after “boot from UART” is triggered.

1. PC sends loader image to BMC.
2. MCU ROM verifies loader image with its signature if secure boot is enabled. Then, MCU ROM releases CPU reset and CPU executes the content of loader image.
3. The main function of loader image is initializing DRAM, UART, timer and some important HW controllers.

After that, it receives uart_fw image from PC side.

4. Loader verifies uart_fw image by uart_fw image's hash embedded in loader image. If uart_fw image is correct, CPU will jump to uart_fw image and execute it.
5. uart_fw implements handshake with PC and get/transfer images from/to PC if user wants to implement spi-flash write/backup process.

Notice:

- There is not loader image for AST1030, AST1035 and AST1060, thus, steps from 1 to 3 can be skipped. uart_fw image is verified directly by MCU ROM.

Chapter 3. Support Environments

- Windows X86-64 (after Win7)
- Linux X86-64 (ASPEED has verified uart_fw on Ubuntu, Redhat and CentOS)

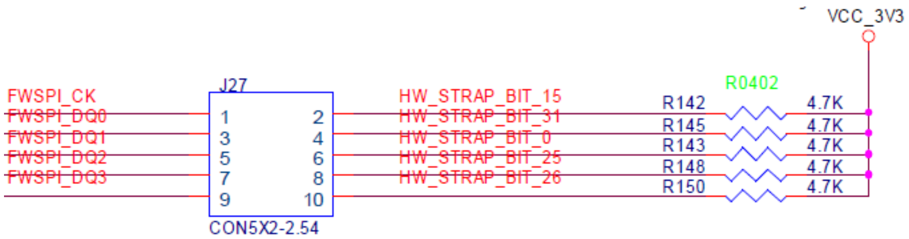
Chapter 4. Board or Pin Settings

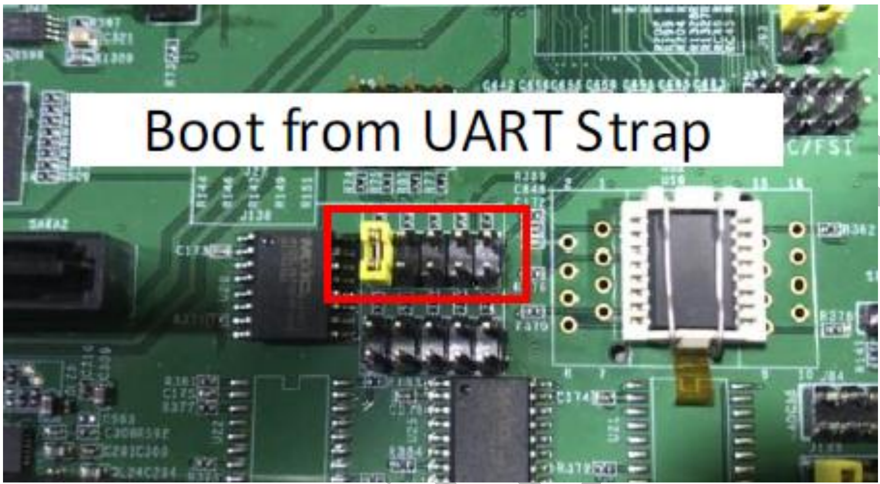
Please enable boot from UART external strap before executing uart_fw_py / uart_fw_py_gui.

Please remember to remove external strap after uart_fw finishes all read/erase/write operations.

4.1 AST2600/AST2620

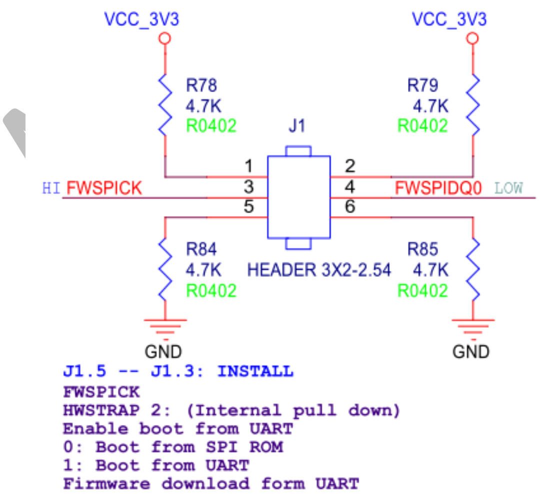
Please connect FWSPICK to VCC_3V3, namely, connect J27 pin 1 and pin 2 on EVB.

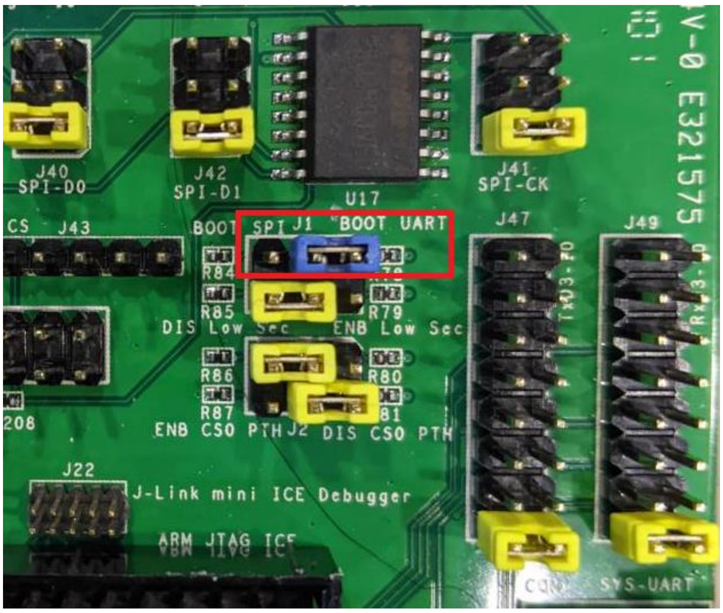




4.2 AST1030/AST1035

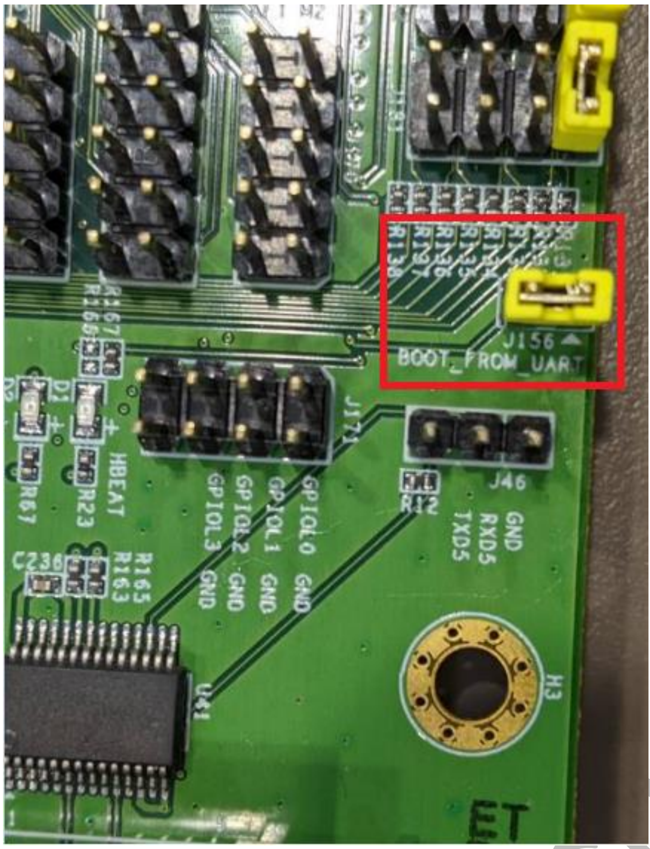
Please connect FWSPICK to VCC_3V3, namely, connect J1 pin 1 and pin 3 on EVB.





4.3 AST1060

Please connect FWSPICK to VCC_3V3, namely, connect J156 pin 1 and pin 2 on EVB.

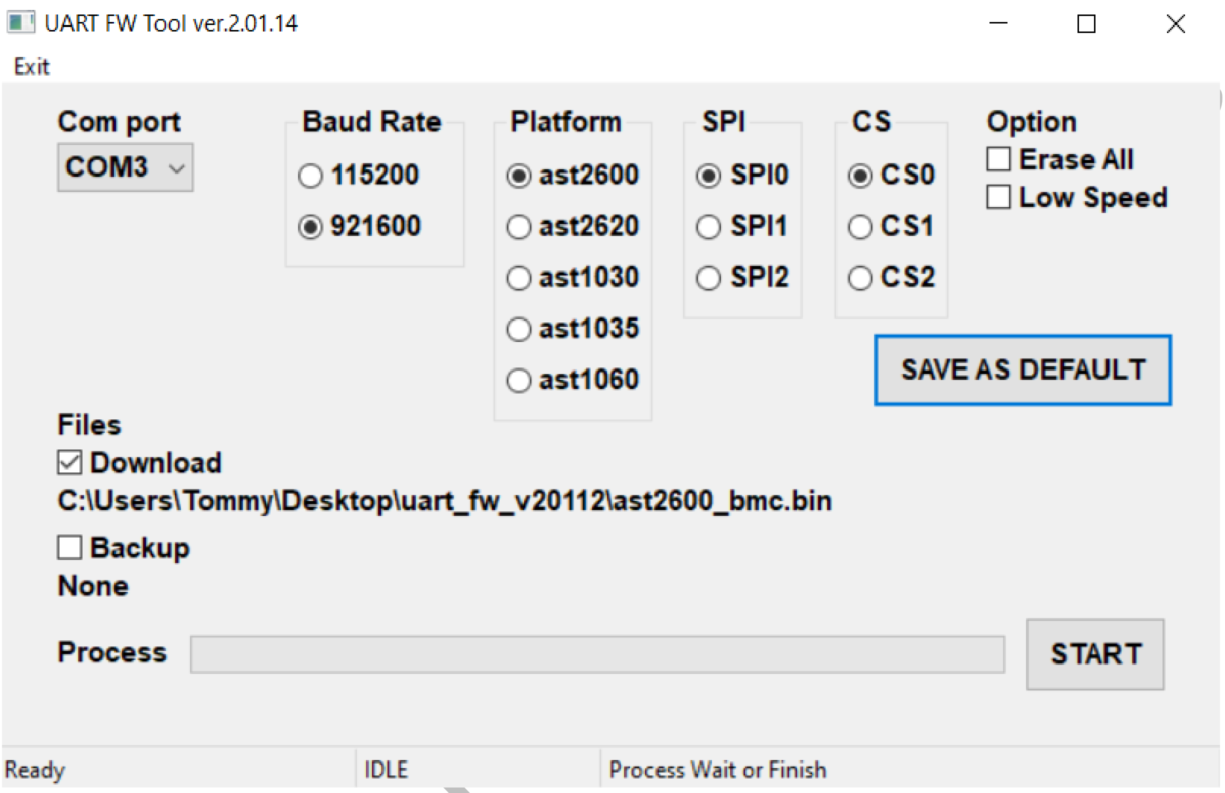


Chapter 5. GUI:

Below are the uart_fw_py GUI version (uart_fw_py_gui) usage that is simple how to use uart_fw.

5.1 GUI layout (Windows version)

Execute the tool by uart_fw_py_gui.exe with administrator right. The tool GUI will be looked like below figure.



- Com port: Select the com port that is connected into BMC.
- Baud Rate: Select the com port baud rate. It could be selected 115200 or 921600 depend on the com port device.
- Platform: Select the BMC product. It could be selected ast2600, ast2620, ast1030, ast1035 and ast1060 in this version.
- SPI: Select SPI controller. It could be selected SPI0(FMC) / SPI1 and SPI2.
- CS: Select chip select. The limit of number is listed below. This item will be limited by HW designed.
  - For baudrate 921600, please check that RS232 adapters on both mother board and PC can support it. Otherwise, uart_fw utility will be stopped abnormally.

  Following lists the supported maximum chip select for each platform.

	FMC	SPI1	SPI2
AST2600/AST2620	3	2	3
AST1030/AST1035	2	2	2
AST1060	2	2	2

- Option: Select other option but necessary function. These items might not be selected in normal case:
  - Erase All: If the item is selected, the whole SPI flash will be erased.
  - Low Speed: If the item is selected, the SPI controller speed will be set into 12.5MHz.
- Files: Select the file that is wanted to download (flash) into BMC flash or backup from BMC flash.
  - Download: If the item is selected, the file selected diagram will be pop up. The file that is wanted to be downloaded into BMC flash should be selected. And the base and length that user wanted to download into BMC flash should be set. If these two items were be set 0x0, the tool will download the file from base 0x0 with length of file size.
  - Backup: If the item is selected, the file selected diagram will be pop up. The file that is wanted to be backed up from BMC flash should be set. And the base and length that user wanted to back up from BMC flash should be set. If the length of file is set 0x0, the tool would back up nothing from BMC flash.
- SAVE AS DEFAULT: The selection of Baud Rate / Platform / SPI / CS and Option will be saved as default value when user click this button. The saved selection will be applied on last time executed.
- START: Start to execute by user setting.

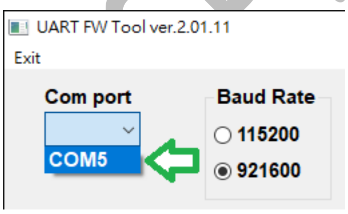
5.2 GUI Examples

The following steps are a sample scenario for usage of uart_fw_py_gui with below setting.

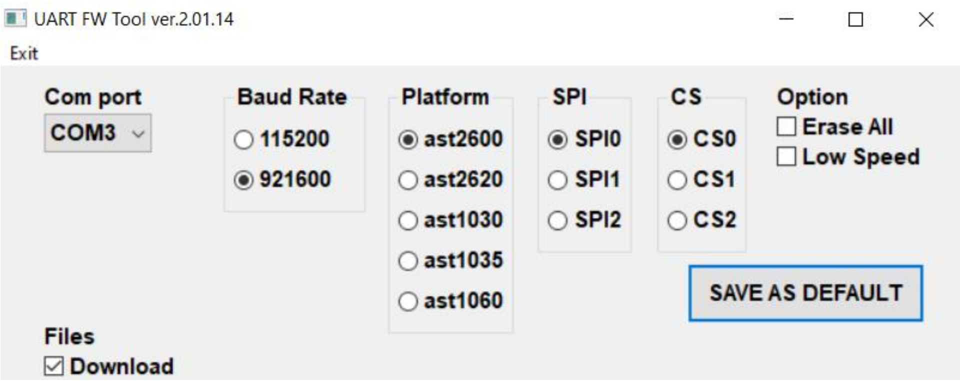
- Com port: COM5 with 921600 baud rate.
- Platform: AST2600 A3
- SPI: SPI0 (FMC)
- CS: CS0
- Option: none
- Download file: (file location)/all.bin with file size at offset 0x0.
- Back up file: (file location)/back.bin with file size 0x200 at offset 0x100.

Step 1: Executed uart_fw_py_gui.exe.

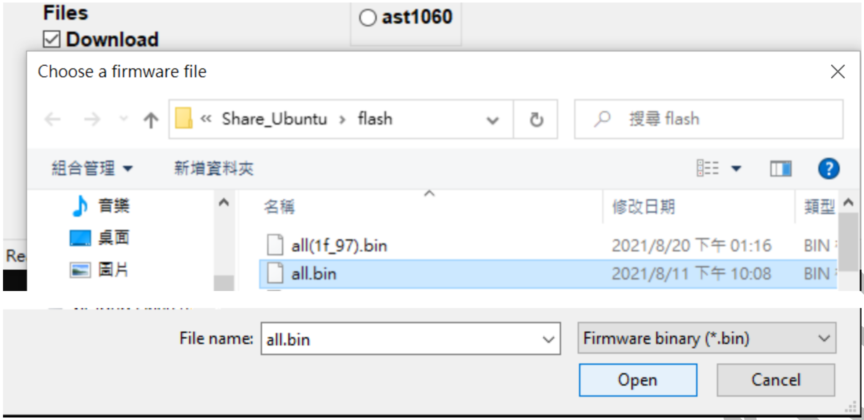
Step 2: Select the com port and baud rate.



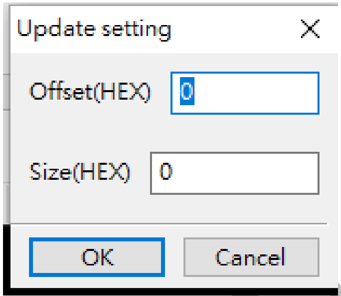
Step 3: Select every item by below setting



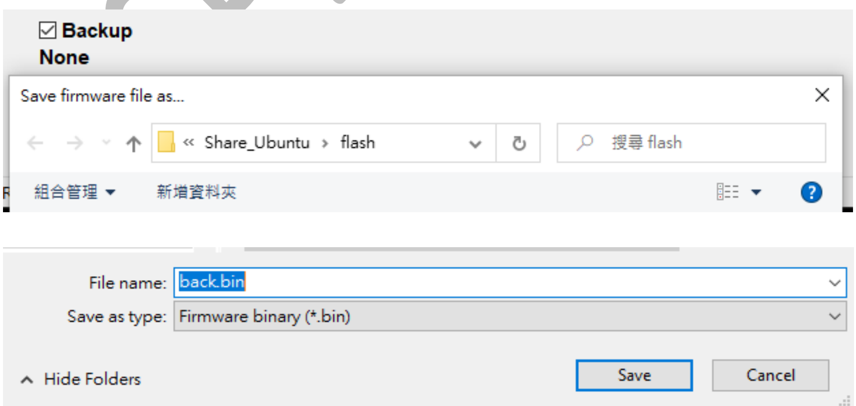
Step 4: Select the file that want to download into BMC flash.



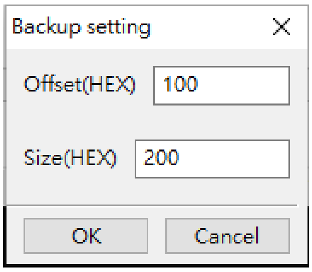
After click Open button, the file size dialog will pop up. Then the download offset and length could be applied here. In this case, the offset and length will be set 0x0.



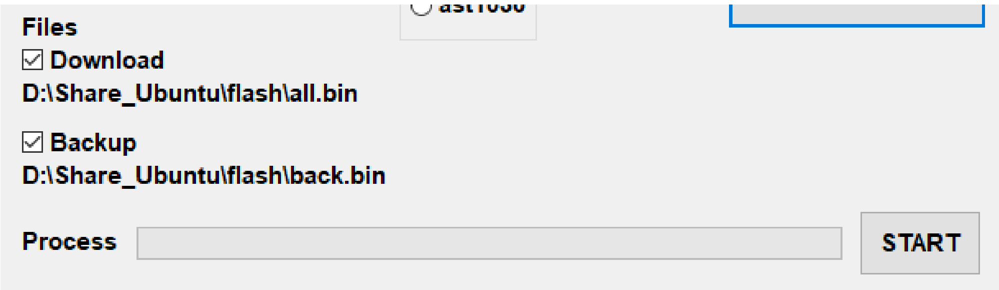
Step 5: Select the file that want to download into BMC flash.



After click Save button, the file size dialog will pop up. Then the backup offset and length could be applied here. In this case, the offset will be set as 0x100 and length will be set 0x200.

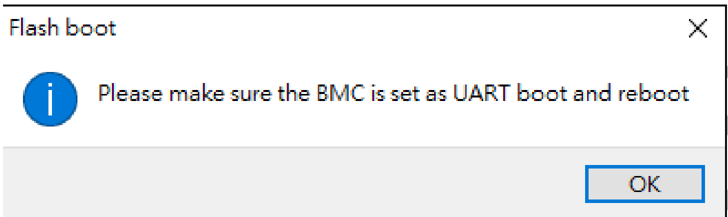


After setting successfully, the file that is download or backup will be shown on the GUI.

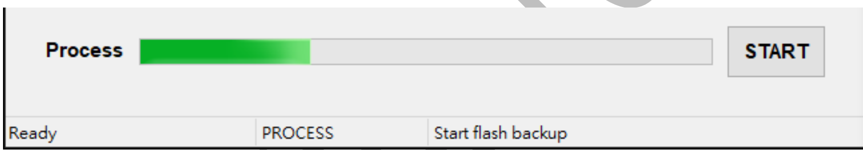


Step 6: Click START button to execute tool.

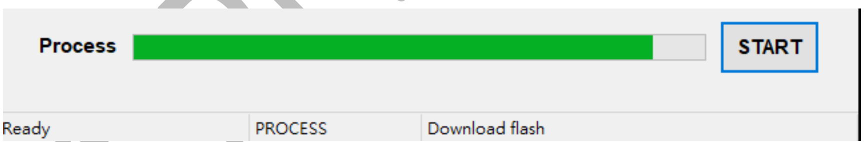
After click START button, the information dialog will pop up. Before you click OK button, **PLEASE MAKE SURE YOUR BMC PLATFORM IS SET BOOTING FROM UART AND RE-BOOT DONE.**

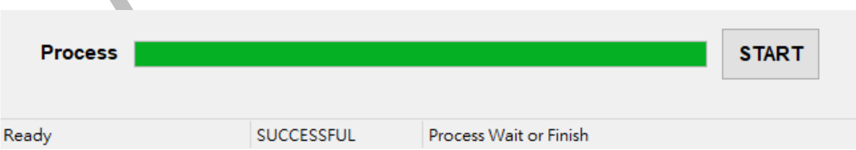


Step 6: Wait the file is downloaded / back up with BMC flash.



The tool status will be updated in the status bar.





5.3 GUI tool under Linux

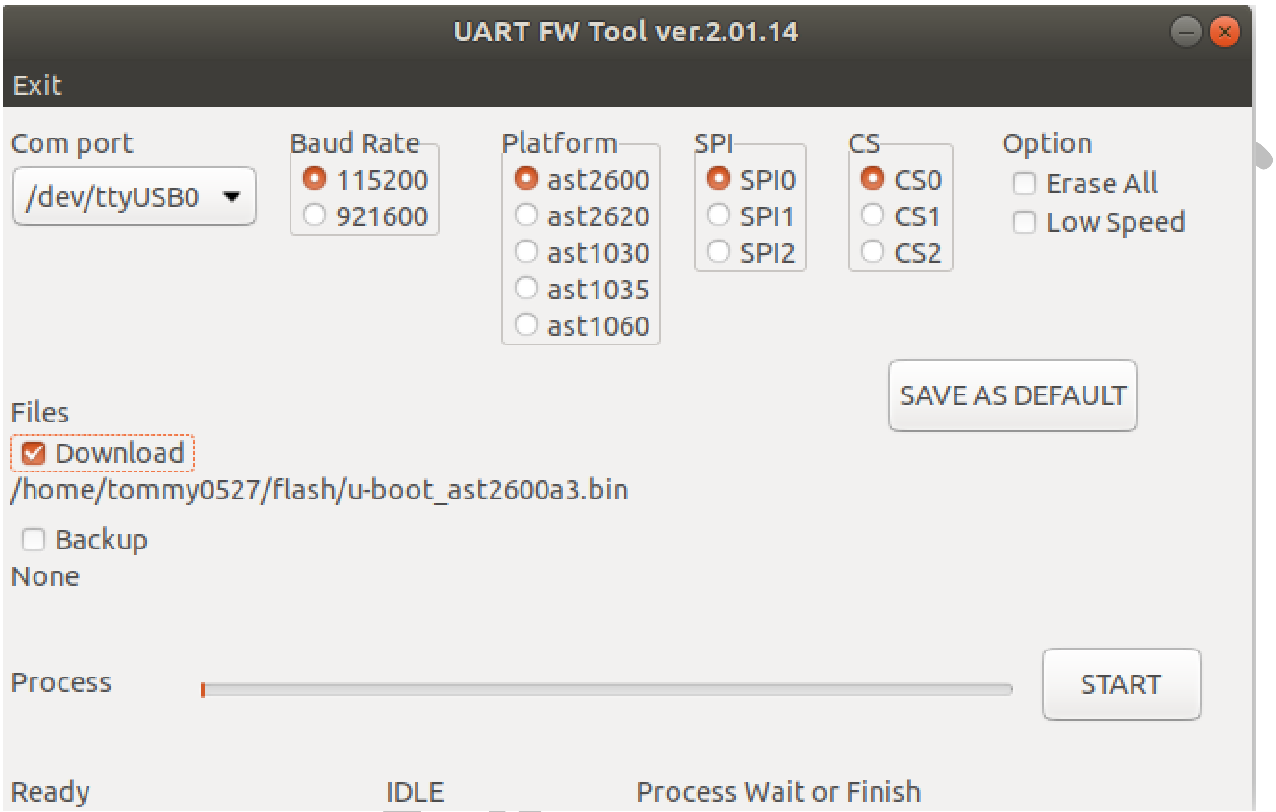
We also provide Linux version tool for user. Because the tool need root right to obtain the information of com port and control it. Please the tool should be executed with root right.

Ex.

Sudo .\uart_fw_py_gui

NOTE!! For now, the GUI tool just be support under Ubuntu 18.04 and 20.04. The other Linux distribution could be support with uart_fw_py command mode that is explained in the chapter 6.

5.4 GUI layout (Linux version)



The tool usage is same with windows version. Please refer it.

Below are command options that is used under command version uart_fw utility.

Chapter 6. Command Line:

6.1 Usage

--platform <aspeed platform>

ASPEED platform.

Support platform: ast2600, ast2620, ast1030, ast1035 or ast1060.

--comport <port name>

Comport used to access BMC. Usually, on Windows, "COM<num>" is used and on Linux "/dev/tty<num>" or "/dev/ttyUSB<num>" is used.

--spi <spi_type> (decimal format)

There are three types of SPI controller, FMC, SPI1 and SPI2.

spi_type: '0' => FMC, '1' => SPI1, '2' => SPI2

--cs <chip select> (decimal format)

Typically, there are 3 chip selections for FMC, 2 for SPI1 and 3 for SPI2. Which depends on the platform. Following lists the supported maximum chip select for each platform.

	FMC	**SPI1**	**SPI2**
AST2600/AST2620	3	2	3
AST1030/AST1035	2	2	2
AST1060	2	2	2

--baudrate <uart baudrate> (decimal format)

Select baudrate for image backup or update. 115200 (default) or 921600.

For baudrate 921600, please check that RS232 adapters on both mother board and PC can support it. Otherwise, uart_fw utility will be stopped after "re-connected to EVB" log.

--input_image <file name>

Image which will be downloaded/updated into flash. By default, 'all.bin' in the same folder will be adopted.

--update_offset <flash_offset> (hex format)

The flash offset where the image is written into. Should be sector size aligned. The default value is zero.

--update_size <size> (hex format)

Write <size> bytes from the start of the input file into flash.

--backup_file <backup file name>

Readback flash content and store it in <backup file>. By default, backup file name is 'backup.bin'

--backup_offset <flash_offset> (hex format)

The flash offset where the image is readback from. The default value is zero.

--backup_size <size> (hex format)

Readback <size> bytes started from the backup offset. If this parameter is not assigned, namely, backup size is zero, the backup process will not be executed.

--erase_all

Erase whole flash. Erase behavior will be executed after backup process if backup size is assigned.

--low_speed

By default, uart_fw use dual mode with 50MHz SPI clock to access SPI flash. Add this parameter will decrease SPI clock to 12.5MHz.

6.2 Command Examples

uart_fw_py.exe is for Windows environment.
uart_fw _py is for Linux environment.

6.2.1 Windows:

- **Please use administrator permission, otherwise, system may stop the execution of uart_fw.**

Secnario 1:

update all.bin image with baudrate 921600 on AST2600 FMC CS0 through COM4 comport

cmd:

uart_fw_py.exe --platform ast2600 --spi 0 --cs 0 --comport COM4 --baudrate 921600 --input_image all.bin

Secnario 2:

backup AST2600 SPI1 CS1 flash content from offset 0x10000 with size 0x200 to file backup_test.bin through COM4 comport

cmd:

uart_fw_py.exe --platform ast2600 --spi 1 --cs 1 --comport COM4 --backup_file backup_test.bin --backup_offset 0x10000 --backup_size 0x200

Secnario 3:

backup AST1030 SPI2 CS0 flash content from offset 0x20000 with size 0x100 to file backup_test.bin through COM4 comport, and erase whole flash, then update all.bin into that flash.

cmd:

uart_fw_py.exe --platform ast1030 --spi 2 --cs 0 --comport COM4 --backup_file backup_test.bin --backup_offset 0x20000 --backup_size 0x100 --erase_all --input_image all.bin

6.2.2 Linux:

- **Remember to change uart_fw_py mode**

cmd: chmod 777 uart_fw_py

Secnario 1:

update all.bin image with baudrate 921600 on AST2600 FMC CS0 through /dev/ttyusb0 comport

cmd:

./uart_fw_py --platform ast2600 --spi 0 --cs 0 --comport /dev/ttyusb0 --baudrate 921600 --input_image all.bin

Secnario 2:

backup AST2600 SPI1 CS1 flash content from offset 0x10000 with size 0x200 to file backup_test.bin through /dev/ttyusb0 comport

cmd:

./uart_fw_py --platform ast2600 --spi 1 --cs 1 --comport /dev/ttyusb0 --backup_file backup_test.bin --backup_offset 0x10000 --backup_size 0x200

Secnario 3:

backup AST1030 SPI2 CS0 flash content from offset 0x20000 with size 0x100 to file backup_test.bin through /dev/ttyusb0 comport, and erase whole flash, then update all.bin into that flash.

cmd:

./uart_fw_py --platform ast1030 --spi 2 --cs 0 --comport /dev/ttyusb0 --backup_file backup_test.bin --backup_offset 0x20000 --backup_size 0x100 --erase_all --input_image all.bin

6.3 Practice Steps and Example

The following are the steps for implementing uart_fw utility. Here, we take AST2600 EVB for example.

Step 1: Power off the device.

Step 2: Connect UART cable to BMC UART. (Turn off the BMC console if it is turned on)

Step 3: Enable “boot from UART” external strap. Please refer to “Board or Pin Settings” section.

Step 4: Execute uart_fw utility command.

Step 5: Power on the device. (This step should be executed after above step)

Step 6: uart_fw implements flash read, erase, or write operation.

Step 7: Remove “boot from UART” external strap and AC on/off the device.

If you use AST2600-A1, please refer to the following steps.

Step 1: Power off the device.

Step 2: Connect UART cable to BMC UART. (Turn off BMC console if it is turned on)

Step 3: Enable “boot from UART” external strap. Please refer to “Board or Pin Settings” section.

Step 4: Execute uart_fw utility command.

Step 5: Power on the device. (This step should be executed after above step)

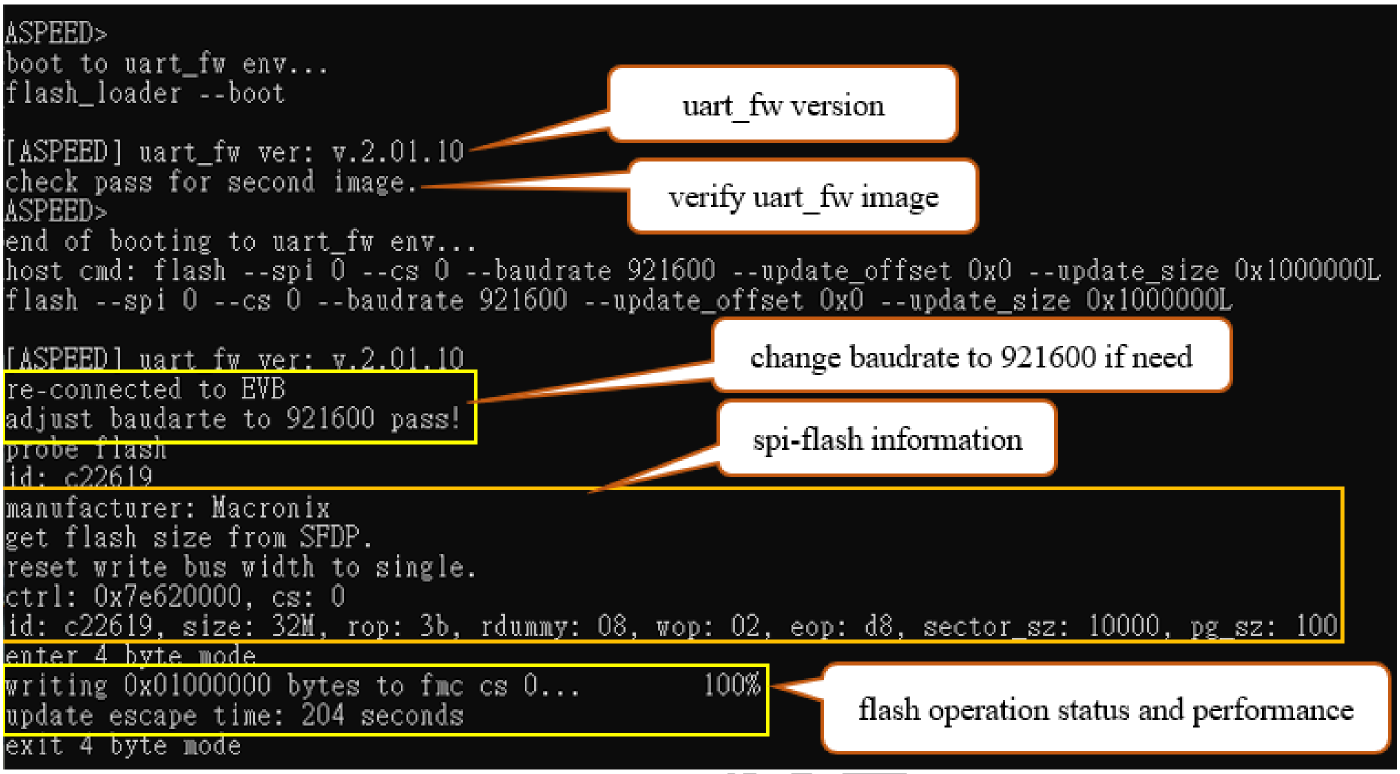
Step 6: Press Enter key when the following message is shown.

```
uart_fw host version: 2.01.10
[info] skip backup flash
connected to EVB
  [Time out] Press 'Enter' to continue...
```

Step 7: uart_fw implements flash read, erase, or write operation.

Step 8: Remove “boot from UART” external strap and AC on/off the device.

6.4 User Interface



Chapter 7. uart_fw - Secure Boot Enabled

If secure boot is enabled, loader image should be signed with the key enabled on OTP.
User can download/install the image signing tool from: https://github.com/AspeedTech-BMC/socsec
Please refer to the "setting up step" section in the above link for signing environment setup.
Before signing the image, original boot from UART header of loader image should be removed by boot_from_uart_hdr_py utility.
After signing the image, a boot from UART header should be added at the beginning of that image.

7.1 Examples:

7.1.1 AST2600/AST2620 Series

If we want to sign ast2600 loader image with test_oem_dss_private_key_2048_1.pem private key by RSA2048 and SHA256 algorithm.

Step 1: Remove boot from header from the beginning of original loader image.

cmd: ./boot_from_uart_hdr_py --input_image ast2600_loader.bin --remove_header

Step 2: Sign image processed after step 1.

cmd: socsec make_secure_bl1_image --soc 2600 --bl1_image ast2600_loader.bin --rsa_sign_key test_oem_dss_private_key_2048_1.pem --output ast2600_loader_out.bin --algorithm RSA2048_SHA256

Step 3: Add boot from UART header for the output image of step 2.

cmd: ./boot_from_uart_hdr_py --input_image ast2600_loader_out.bin --add_header

Step 4: Rename ast2600_loader_out.bin to ast2600_loader.bin. Put ast2600_loader.bin with signature and boot from UART header to the uart_fw utility folder.

7.1.2 AST1030/AST1035/AST1060

There is no loader image for AST1030, AST1035 and AST1060. Thus, our signing image is uart_fw image itself.

If we want to sign ast1030, ast1035 or ast1060 uart_fw image with test_oem_dss_private_key_2048_1.pem private key by RSA2048 and SHA256 algorithm.

Step 1: Remove boot from header from the beginning of original uart_fw image.

cmd: ./boot_from_uart_hdr_py --input_image ast1030_uart_fw.bin --remove_header

Step 2: Sign image processed after step 1.

cmd: socsec make_secure_bl1_image --soc 1030 --bl1_image ast1030_uart_fw.bin --rsa_sign_key test_oem_dss_private_key_2048_1.pem --output ast1030_uart_fw_out.bin --algorithm RSA2048_SHA256

Step 3: Add boot from UART header for the output image of step 2.

cmd: ./boot_from_uart_hdr_py --input_image ast1030_uart_fw_out.bin --add_header

Step 4: Rename ast1030_uart_fw_out.bin to ast1030_uart_fw.bin. Put ast1030_uart_fw.bin with signature and boot from UART header to the uart_fw utility folder.

Chapter 8. Supported SPI Flash Parts

The following SPI flash parts have been verified on ASPEED EVB with uart_fw utility.

Normally, if a SPI flash has uniform erase sector size by default and without HW problem about SFDP table, it can be supported by uart_fw utility by default. User can try to read/write flash before asking ASPEED whether current flash is supported by uart_fw utility.

Vendor	MPN	ID
Micron	MT25QL128ABA8ESF	0x20BA18
	MT25QL256ABA8ESF	0x20BA19
	MT25QL512ABB8ESF	0x20BA20
	MT25QL01GBBB8ESF	0x20BA21
Macronix	MX25L12835FMI	0xC22018
	MX25L25635FMI	0xC22019
	MX66L51235FMI	0xC2201A
	MX25L51245GMI	0xC2201A

	MX66L1G45GMI	0xC2201B
	MX66L2G45G	0xc2201c
Winbond	W25Q80DV	0xEF4014
	W25Q256JVFIQ	0xEF4019
	W25Q256JVFM	0xEF7019
	W25Q512JVFIQ	0xEF4020
	W25Q512JVFM	0xEF7020
	W25Q01JVFIQ	0xEF4021
	W25Q01JVFIM	0xEF7021
	W25Q02JVIM	0xEF7022
GigaDevice	GD25Q256D	0xC84019
	GD25B512M	0xC8471A
	GD55B512M	0xC8401A
	GD55B01GE	0xC8471B
	GD55B02GE	0xC8471C
Cypress	S25FL256L	0x016019
	S25HL512T	0x342A1A
	S25HL01GT	0x342A1B
ISSI	IS25LP256	0x9D6019
	IS25LP512M	0x9D601A
	IS25LP01G	0x9D601B
BIG INNOVATION	B25B512M	0xC8471A
	B55B01G	0xC8471B
	B55B02G	0xC8471C

Chapter 9. Notice

- For baudrate 921600 capability, it depends on the ability of RS232 UART adapters on both BMC and PC sites.
- If you execute uart_fw on ASPEED EVB and the adapter on your PC does not supported baudrate 921600, you can use an extra USB adapter to connect EVB UART port. Following USB-UART cable has been verified by ASPEED internally. http://www.sunbox.com.tw/English/Products/Detail/40
- Please turn off your original BMC console terminal before executing uart_fw_py since it transfers image through UART path.
- Normally, since uart_fw_py utility uses "boot from UART" feature, you may power on device after executing the command. Otherwise, uart_fw utility will be timeout when waiting 'boot from UART flag.
- For AST2600-A1, since there is a HW problem about MCU ROM UART message, user need to press "Enter" key after timeout.

```
uart_fw host version: 2.01.10
[info] skip backup flash
connected to EVB
  [Time out] Press 'Enter' to continue...
```

Chapter 10. Release Note

Version	Author	Description
2.01.10	Chin-Ting Kuo	- First official release. - Support AST2600, AST2605 and AST1030 platforms.
2.01.11	Chin-Ting Kuo	- [AST2600] [AST2605] [AST1030] Fix bug when the updated image size is not multiple of 4.
	Tommy Huang	- Windows/Linux GUI support.
2.01.12	Chin-Ting Kuo	- AST1060 support.
2.01.13	Tommy Huang	- GUI: Fix CS0 and CS1 selection problem.
2.01.14	Chin-Ting Kuo	- AST1030: Boot from UART1 support. - AST2600/AST2620: Support S25HL512 and S25HL01G flash parts. - Remove AST2605 platform series. - AST1035 platform support.